Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The correct sequence which shows decreasing order of the ionic radii of the elements is

(A) $Al^{3+} > Mg^{2+} > Na^{+} > F^{-} > O^{2-}$

(B) $Na^{+} > Mg^{2+} > Al^{3+} > O^{2-} > F^{-}$

(C) $Na^{+} > F^{-} > Mg^{2+} > O^{2-} > Al^{3+}$

(D) $O^{2-} > F^{-} > Na^{+} > Mg^{2+} > Al^{3+}$

Q.2

Which is not the correct order for the stated property.

(A) $Ba > Sr > Mg$; atomic radius

(B) $F > O > N$; first ionization enthalpy

(C) $Cl > F > I$; electron affinity

(D) $O > Se >$ Te ; electronegativity

Q.3

Which of the following order is wrong?

(A) $NH_3 < PH_3 < AsH_3 -$ Acidic

(B) $Li < Be < B < C -$ First IP

(C) $Al_2O_3 < MgO < Na_2O < K_2O -$ Basic

(D) $Li^{+} < Na^{+} < K^{+} < Cs^{+}$ - Ionic radius

Q.4

Identify the correct order of the size of the following:

(A) $Ca^{2+} < K^{+} < Ar < Cl^{-} < S^{2-}$

(B) $Ar < Ca^{2+} < K^{+} < Cl^{-} < S^{2-}$

(C) $Ca^{2+} < Ar < K^{+} < Cl^{-} < S^{2-}$

(D) $Ca^{2+} < K^{+} < Ar < S^{2-} < Cl^{-}$

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Q.5

$\frac{N_0}{2}$ atoms of Xg are converted into X^+ g by energy E_1 . $\frac{N_0}{2}$ atoms of Xg are converted into X^-g by energy E_2 . , then ionization potential and electron affinity of Xg respectively are

(A) $\frac{2E_1}{N_0}, \frac{2(E_1-E_2)}{N_0}$

(B) $\frac{2E_1}{N_0}, \frac{2E_2}{N_0}$

(C) $\frac{(E_1-E_2)}{N_0}, \frac{2E_2}{N_0}$

(D) None is correct

Q.6

Which of the following elements represents highly electropositive as well as highly electronegative character in its period?

(A) Hydrogen

(B) Nitrogen

(C) Fluorine

(D) None of these

Q.7

Match Column-I with Column-II and select the correct answer by the given codes.

Columnn-I (Atoms)		**Column-II (Properties)**
(A) He	(p)	High electronegative
(B) F	(q)	Most electropositive
(C) Rb	(r)	Strongest reducing agent
(D) Li	(s)	Highest ionisation energy

(A) $A-(s), B-(q), C-(r), D-(p)$

(B) $A-(p), B-(s), C-(q), D-(r)$

(C) $A-(s), B-(p), C-(r), D-(q)$

(D) $A-(s), B-(p), C-(q), D-(r)$

Q.8

Na^+, Mg^{2+}, Al^{3+} and Si^{4+} are isoelectronic. The order of their ionic size is

(A) $Na^+ > Mg^{2+} < Al^{3+} < Si^{4+}$

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(B) $Na^{+} < Mg^{2+} > Al^{3+} > Si^{4+}$

(C) $Na^{+} > Mg^{2+} > Al^{3+} > Si^{4+}$

(D) $Na^{+} < Mg^{2+} > Al^{3+} < Si^{4+}$

Q.9

The set of three elements having successive atomic numbers and having the ionization energies of 2372,520 and $890\ kJ$ per mol is

(A) H, He, Li

(B) He, Li, Be

(C) Li, Be, B

(D) B, C, N

Q.10

In a given energy level, the order of penetration effect of different orbitals is

(A) $f < d < p < s$

(B) $s = p = d = f$

(C) $s < p < d < f$

(D) $p > s > d > f$

Q.11

What is the order of ionisation energies of the coinage metal

(A) $Cu > Ag < Au$

(B) $Cu > Ag > Au$

(C) $Cu < Ag < Au$

(D) $Ag > Au < Cu$

Q.12

What is the correct order of electronegativity?

(A) $M^{1-} < M^{2-} < M^{3-} < M^{4-}$

(B) $M^{1-} > M^{2-} > M^{3-} > M^{4-}$

(C) $M^{1-} < M^{2-} < M^{3-} < M^{4-}$

(D) $M^{4-} < M^{2-} < M^{3-} < M^{1-}$

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Q.13

In which of the following process highest energy is absorbed?

(A) $\mathrm{Cu} \rightarrow \mathrm{Cu}^{+}$

(B) $\mathrm{Br} \rightarrow \mathrm{Br}^{-}$

(C) $\mathrm{I} \rightarrow \mathrm{I}^{-}$

(D) $\mathrm{Li} \rightarrow \mathrm{Li}^{+}$

Q.14

Which of the following arrangements represents the increasing order (smallest to largest) of ionic radii of the given species $\mathrm{O}^{2-}, \mathrm{S}^{2-}, \mathrm{N}^{3-}, \mathrm{P}^{3-}$?

(A) $\mathrm{O}^{2-}<\mathrm{N}^{3-}<\mathrm{S}^{2-}<\mathrm{P}^{3-}$

(B) $\mathrm{O}^{2-}<\mathrm{P}^{3-}<\mathrm{N}^{3-}<\mathrm{S}^{2-}$

(C) $\mathrm{N}^{3}<\mathrm{O}^{2-}<\mathrm{P}^{3-}<\mathrm{S}^{2-}$

(D) $\mathrm{N}^{3-}<\mathrm{S}^{2-}<\mathrm{O}^{2-}<\mathrm{P}^{3-}$

Q.15

The incorrect statement among the following is

(A) the first ionization potential of Al is less than the first ionization potential of Mg

(B) the second ionization potential of Mg is greater than the second ionization potential of Na

(C) the first ionization potential of Na is less than the first ionization potential of Mg

(D) the third ionization potential of Mg is greater than the third ionization potential of Al.

Q.16

The electronic contigurations of four elements are given below. Arrange these elements in the correct order of magnitude of their electron affinity

(i) $2\ \mathrm{s}^{2}2\mathrm{p}^{5}$

(ii) $3\ \mathrm{s}^{2}3\mathrm{p}^{5}$

(iii) $2\ \mathrm{s}^{2}2\mathrm{p}^{4}$

(iv) $3\ \mathrm{s}^{2}3\mathrm{p}^{4}$

Select the correct answer using the codes

(A) (i) $<(\mathrm{ii})<(\mathrm{iii})<(\mathrm{iv})$

(B) (ii) $<(\mathrm{i})<(\mathrm{iv})<(\mathrm{iii})$

(C) (iii) $<$ (iv) $<(\mathrm{i})<(\mathrm{ii})$

(D) (iii) $<$ (iv) $<(\mathrm{ii})<(\mathrm{i})$

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Q.17

The alkali metal that reacts with nitrogen directly to fom nitride is

(A) Li

(B) Na

(C) K

(D) $\mathbf{Rb}$

Q.18

The decreasing order of the second ionization potential of K, Ca and Ba

(A) $\mathrm{K} > \mathrm{Ca} > \mathrm{Ba}$

(B) $\mathrm{Ca} > \mathrm{Ba} > \mathrm{K}$

(C) $\mathrm{Ba} > \mathrm{K} > \mathrm{Ca}$

(D) $\mathbf{K} > \mathrm{Ba} > \mathrm{Ca}$

Q.19

Which of the following is most volatile compound?

(A) HI

(B) HCl

(C) HBr

(D) HF

Q.20

Which among the following elements of 3nd period shows maximum tendency of forming $\mathrm{d}\pi - \mathrm{p}\pi$ bond?

(A) Chlorine

(B) Silicon

(C) Phosphorus

(D) Sulphur

Answers & Solutions

Q.1 Answer:

$\mathrm{O}^{2-} > \mathrm{F}^{-} > \mathrm{Na}^{+} > \mathrm{Mg}^{2+} > \mathrm{Al}^{3+}$

Solution:

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All the given species contains $10e^-$ each i.e. isoelectronic. For isoelectronic species anion having high negative charge is largest in size and the cation having high positive charge is smallest.

Q.2 Answer:

$F > O > N$; first ionization enthalpy

Solution:

On moving along the period, ionization enthalpy increases. In second period, the order of ionization enthalpy should be as follows:

$F > O > N$

But N has half-filled structure, therefore, it is more stable than O. That is why its ionization enthalpy is higher than O. Thus, the correct order of IE is

$F > N > O$

Q.3 Answer:

$Li < Be < B < C-$ First IP

Solution:

Along the period, I.P. generally increases but not regularly. Be and B are exceptions. First I.P. increases in moving from left to right in a period, but I.P. of B is lower than Be.

Q.4 Answer:

$Ca^{2+} < K^+ < Ar < Cl^- < S^{2-}$

Solution:

For isoelectronic species, size of anion increases as negative charge increases whereas size of cation decreases with increase in positive charge. Further ionic radii of anions is more than that of cations. Thus the correct order is $Ca^{++} < K^+ < Ar < Cl^- < S^{--}$

Q.5 Answer:

$\frac{2E_1}{N_0}, \frac{2E_2}{N_0}$

Solution:

$Xg \longrightarrow X^+g + e^-$

E_1 **for** $\frac{N_0}{2}$ **atoms**

$\frac{N_0}{2}$ **atoms of** Xg **have been ionized, by energy** $= E_1$

Thus, ionization energy $Xg = \frac{2E_1}{N_0}$ **per atom**

$Xg + e^- \longrightarrow X^-g$

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E_2 **for** $\frac{N_0}{2}$ **atoms**

Thus, electron affinity of Xg **is** $\frac{2E_0}{N_0}$ **per atom.**

Q.6 Answer:

Hydrogen

Solution:

First period has H **and He only out of which He is inert, hence, H behaves as a highly electropositive as well as electronegative.**

Q.7 Answer:

$A-(s), B-(p), C-(q), D-(r)$

Solution:

Helium (He) $1\ s^2 \rightarrow$ **Highest ionisation energy due to noble gas in nature.**
Fluorine (F) $1\ s^2 2\ s^2 2p^3 \rightarrow$ **High electronegativity in nature due to small size and** -1 **oxidation state.**
Rubidium (Rb) $\rightarrow$ **Most electropositive element due to large atomic size.**
Lithium (Li) $\rightarrow$ **Strongest reducing agent due to small size and positive oxidation state** $(+1)$

Q.8 Answer:

$Na^+ > Mg^{2+} > Al^{3+} > Si^{4+}$

Solution:

Among the isoelectronic species, the cation having more positive charge would be the smaller in size.

Q.9 Answer:

He, Li, Be

Solution:

He $(Z=2)$ **is a noble gas and has highest**
I.E. The I.E. of $Be(Z=4)$ **is more than that of** $Li(Z=3)$

Q.10 Answer:

$f < d < p < s$

Solution:

Penetration effect order is $s > p > d > f$.

Q.11 Answer:

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$\mathrm{Cu} > \mathrm{Ag} < \mathrm{Au}$

Solution:

$\mathrm{Cu} = 3d^{10}4s^1, \mathrm{Ag} = 4d^{10}5\ \mathrm{s}^1, \mathrm{Au} = 4f^{14}5d^{10}6s^1$

In all the above given cases, unpaired s -electron has to be removed. In the case of Cu, a $4s$ electron is to be removed which is closer to the nucleus than the $5s$ electron of Ag.

So, IE_1 of $\mathrm{Cu} > \mathrm{IE}_1$ of Ag.

However, in case of Au, due to imperfect screening effect of $14\mathrm{e}^- s$ of $4f$ orbitals, the nuclear charge increases and therefore 5s e^- of Au is more tightly held. Thus, the order of IE_1 is $\mathrm{Cu} > \mathrm{Ag} < \mathrm{Au}$.

Q.12 Answer:

$M^{1-} > M^{2-} > M^{3-} > M^{4-}$

Solution:

Higher the positive charge the greater is the EN and higher the negative charge the lesser is the EN.

So, EN order is $\mathrm{M}^- > \mathrm{M}^{2-} > \mathrm{M}^{3-} > \mathrm{M}^{-4}$

Q.13 Answer:

$\mathrm{Cu} \rightarrow \mathrm{Cu}^+$

Solution:

Cu has completely filled d -orbital, so highest energy is absorbed when it converts to Cu^+ ion.

Q.14 Answer:

$\mathrm{O}^{2-} < \mathrm{N}^{3-} < \mathrm{S}^{2-} < \mathrm{P}^{3-}$

Solution:

For isoelectronic species ionic radin decreases as the charge on ion decreases. Further on moving down in a group ionic radii increases. Hence the correct order is $\mathrm{O}^{2-} < \mathrm{N}^{3-} < \mathrm{S}^{2-} < \mathrm{P}^{3-}$

Q.15 Answer:

the second ionization potential of Mg is greater than the second ionization potential of Na

Solution:

IE_2 of Mg is lower than that of Na because in case of Mg^+, $3s$ -electron has to be removed where as in case of Na^+, an electron is removed from the stable inert gas configuration which is difficult.

Q.16 Answer:

(iii) < (iv) < (i) < (ii)

Solution:

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Electron affinity increases with the decrease in the size and it also depends mon the electronic configuration

Q.17 Answer:

Li

Solution:

$\mathbf{6Li + N_2 \rightarrow 2Li_3N}$ **Lithium nitride**

Q.18 Answer:

$\mathrm{K > Ca > Ba}$

Solution:

In case of alkali metals the second $\mathrm{e^-}$ is to be removed from a cation which has already acquired a noble gas configuration. On the other hand, in the alkaline earth metals the second $\mathrm{e^-}$ is to be removed from a monovalent cation (Cat) which still has $\mathrm{e^-}$ in the outer shell, thus the second e^- can be removed easily $\&$ it becomes more easier in increasing atomic size on down to the group. Hence the correct decreasing order of second LP is

$\mathrm{K > Ca > Ba}$

Q.19 Answer:

HCl

Solution:

Boiling point of HF is highest dne to H-bonding. For other halogen acids b.p. increases in the order $\mathrm{HCl < HBr < HI}$. Therefore, most volatile (with Lower b.pt) is HCl

Q.20 Answer:

Chlorine

Solution:

Among the given elements Cl has maximum tendency of $\mathrm{d\pi - p\pi}$ bonding On moving across period, nuclear charge increases and more s, p-electrons are added. Due to incomplete shielding of nuclear charge, the atomic size and so the size of 3 d-orbitals decreaxes from $\mathrm{Si < P < S < Cl}$

Decrease in size of 3 d-orbitals leads to stronger $d\pi - p\pi$ overlap.

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